Statistics & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Reading data from a bar chart**

The bar chart shows the number of books sold each day at a bookstore over five days.

a Find the total number of books sold over the five days. Total $= 12 + 18 + 9 + 15 + 21 = 75$ books.

b Find the mean number of books sold per day. Mean $= \frac{75}{5} = 15$ books per day.

(a) Total $= 12 + 18 + 9 + 15 + 21 = 75$ books.; (b) Mean $= \frac{75}{5} = 15$ books per day.

2 **[Foundational]** • **Target: Mean, median, and mode**

Find the mean of $4, 8, 6, 10, 7$.

Sum $= 35$. Mean $= \frac{35}{5} = 7$.

3 **[Foundational]** • **Target: Mean, median, and mode**

Find the median of $3, 9, 5, 1, 7$.

Sorted: $1, 3, 5, 7, 9$. Median $= 5$.

4 **[Foundational]** • **Target: Mean, median, and mode**

Find the mode of $2, 4, 4, 6, 8, 4, 2$.

4 appears three times, more than any other value, so the mode is 4.

5 **[Foundational]** • **Target: Mean, median, and mode**

A data set has values $8, 12, 10, 15, 10, 14, 11$. Find the mean, median, and mode.

Sum $= 80$, mean $= \frac{80}{7} \approx 11.43$. Sorted: $8, 10, 10, 11, 12, 14, 15$; median (4th value) $= 11$. Mode $= 10$ (appears twice).

6 **[Foundational]** • **Target: Range and quartiles**

Find the range of $5, 12, 8, 20, 3$.

Range $= 20 - 3 = 17$.

7 **[Foundational]** • **Target: Range and quartiles**

For the data set $2, 4, 5, 7, 8, 9, 12$, find Q_1 and Q_3.

Median (4th value) $= 7$. Lower half $2, 4, 5$: $Q_1 = 4$. Upper half $8, 9, 12$: $Q_3 = 9$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Range and quartiles**

Find the interquartile range (IQR) for the same data set.

$\text{IQR} = Q_3 - Q_1 = 9 - 4 = 5.$

9 **[Intermediate]** • **Target: Variance and standard deviation**

Find the variance of $2, 4, 6, 8, 10$, using $\sigma^2 = \frac{\sum (x-\mu)^2}{n}$.

Mean $= 6$. Squared deviations: $16, 4, 0, 4, 16$, sum $= 40$. Variance $= \frac{40}{5} = 8$.

10 **[Intermediate]** • **Target: Variance and standard deviation**

Find the standard deviation of the same data set, rounded to two decimal places.

$\sigma = \sqrt{8} \approx 2.83.$

11 **[Intermediate]** • **Target: Variance and standard deviation**

A data set has variance 25. State its standard deviation.

$\sigma = \sqrt{25} = 5.$

12 **[Intermediate]** • **Target: Frequency tables**

A frequency table shows value 1 with frequency 3, value 2 with frequency 5, and value 3 with frequency 2. Find the mean.

Sum $= 1(3) + 2(5) + 3(2) = 3 + 10 + 6 = 19$. Total frequency $= 10$. Mean $= \frac{19}{10} = 1.9$.

13 **[Intermediate]** • **Target: Frequency tables**

Using the same frequency table, find the mode.

Value 2 has the highest frequency (5), so the mode is 2.

14 **[Intermediate]** • **Target: Frequency tables**

Using the same frequency table, find the median.

There are 10 values in order $1, 1, 1, 2, 2, 2, 2, 2, 3, 3$. Median $= \frac{2+2}{2} = 2$ (average of the 5th and 6th values).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Basic probability • Cross-Lesson: 8.01 Exponentials & Logarithms

A fair six-sided die is rolled. Find $P(\text{number} > 4)$.

Favorable outcomes $\{5, 6\}$: $P = \frac{2}{6} = \frac{1}{3}$.

16 [Challenging] • Target: Basic probability • Cross-Lesson: 1.01 Algebra & Equations

A bag contains 5 red and 3 blue marbles. Find $P(\text{blue})$.

$P(\text{blue}) = \frac{3}{8}$.

17 [Challenging] • Target: Basic probability • Cross-Lesson: 2.01 Functions & Transformations

A card is drawn from a standard 52-card deck. Find $P(\text{king})$.

$P(\text{king}) = \frac{4}{52} = \frac{1}{13}$.

18 [Challenging] • Target: Basic probability • Cross-Lesson: 3.01 Sequences & Series

Two fair coins are flipped. Find $P(\text{both heads})$.

$P = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

19 [Challenging] • Target: Complementary and mutually exclusive events • Cross-Lesson: 4.01 Trigonometry

If $P(A) = 0.35$, find $P(\text{not } A)$.

$P(\text{not } A) = 1 - 0.35 = 0.65$.

20 [Challenging] • Target: Complementary and mutually exclusive events • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

A die is rolled. Events $A = \{\text{even}\}$ and $B = \{\text{rolling a } 5\}$. Are A and B mutually exclusive?

Yes — $\{2, 4, 6\}$ and $\{5\}$ share no outcomes.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Frequency tables

The bar chart shows a frequency table. Find the total number of data points.

a. 15 ✓

b. 6

c. 3

d. 10

$4 + 6 + 5 = 15$.

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22 [Foundational] • Target: MCQ — Reading data from a bar chart

The bar chart shows the number of books sold each day at a bookstore over five days. Find the total books sold.

a. 87 ✓

b. 22

c. 77

d. 97

$12 + 18 + 15 + 22 + 20 = 87$.

23 [Foundational] • Target: MCQ — Reading data from a bar chart

Using the same bar chart, find the mean number of books sold per day.

a. 15

b. 18

c. 20

d. 17.4 ✓

$87 \div 5 = 17.4$.

24 [Foundational] • Target: MCQ — Reading data from a bar chart

Using the same bar chart, find the range of books sold.

a. 12

b. 22

c. 8

d. 10 ✓

$22 - 12 = 10$.

25 [Foundational] • Target: MCQ — Reading data from a bar chart

Using the same bar chart, on which day were the most books sold?

a. Wednesday

b. **Thursday** ✓

c. Tuesday

d. Friday

Thursday shows the tallest bar, 22 books.

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26 [Foundational] • Target: MCQ — Mean, median, and mode

Find the mean of $5, 9, 7, 11, 8$.

a. 8 ✓

b. 7

c. 40

d. 9

Sum $= 40$; mean $= 40 \div 5 = 8$.

27 [Foundational] • Target: MCQ — Mean, median, and mode

Find the mean of $12, 18, 15, 21, 14$.

a. 15

b. 16 ✓

c. 80

d. 18

Sum $= 80$; mean $= 80 \div 5 = 16$.

28 [Foundational] • Target: MCQ — Variance and standard deviation

A data set has variance 36. State its standard deviation.

a. 6 ✓

b. 12

c. 18

d. $1{,}296$

$\sqrt{36} = 6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Basic probability

A fair six-sided die is rolled. Find $P(\text{number} < 3)$.

a. $\frac{1}{6}$

b. $\frac{1}{2}$

c. $\frac{1}{3}$ ✓

d. $\frac{2}{3}$

$\{1, 2\}$: $\frac{2}{6} = \frac{1}{3}$.

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30 [Intermediate] • Target: MCQ — Complementary and mutually exclusive events

If $P(A) = 0.28$, find $P(\text{not } A)$.

a. 0.28

b. 1.28

c. 0.5

d. 0.72 ✓

$1 - 0.28 = 0.72$.

31 [Intermediate] • Target: MCQ — Independent events

A fair coin is flipped twice. Find $P(\text{both heads})$.

a. $\frac{1}{3}$

b. 1

c. $\frac{1}{4}$ ✓

d. $\frac{1}{2}$

$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

32 [Intermediate] • Target: MCQ — Conditional probability

A box has 12 balls: 7 red and 5 blue. Two balls are drawn without replacement. Given the first is blue, find $P(\text{second is blue})$.

a. $\frac{5}{11}$

b. $\frac{4}{11}$ ✓

c. $\frac{5}{12}$

d. $\frac{4}{12}$

After removing one blue, 4 blue remain out of 11 total.

33 [Intermediate] • Target: MCQ — Counting: permutations and combinations

A code uses 4 distinct digits from 0–9, where order matters. How many possible codes are there?

a. 10,000

b. 40

c. 210

d. 5,040 ✓

$10 \times 9 \times 8 \times 7 = 5{,}040$.

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34 [Intermediate] • Target: MCQ — Word problems: statistics and probability in context

A survey of 250 students found that 150 like math and 100 like science, with 40 liking both. Find the probability that a randomly selected student likes math or science.

a. 0.84 ✓

b. 0.6

c. 0.16

d. 1.0

$\frac{150+100-40}{250} = \frac{210}{250} = 0.84$.

35 [Intermediate] • Target: MCQ — Variance and standard deviation

Find the mean of $2, 6, 1, 13, 8$.

a. 30

b. 6 ✓

c. 7

d. 5

Sum $= 30$. Mean $= \frac{30}{5} = 6$.

36 [Intermediate] • Target: MCQ — Frequency tables

A bag contains 6 red and 3 blue marbles. Find $P(\text{blue})$.

a. $\frac{3}{11}$

b. $\frac{2}{3}$

c. $\frac{4}{9}$

d. $\frac{1}{3}$ ✓

$P(\text{blue}) = \frac{3}{9} = \frac{1}{3}$.

37 [Intermediate] • Target: MCQ — Basic probability

Find the range of $28, 2, 14, 37, 32$.

a. 35 ✓

b. 37

c. 38

d. 33

Range $= 37 - 2 = 35$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: MCQ — Complementary and mutually exclusive events • Cross-Lesson: 3.01 Sequences & Series

Find the median of $2, 19, 28, 15, 50$.

a. 19 ✓

b. 22

c. 17

d. 21

Sorted: $2, 15, 19, 28, 50$. Median $= 19$.

39 [Challenging] • Target: MCQ — Independent events • Cross-Lesson: 4.01 Trigonometry

Two independent events have $P(A) = 0.7$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

a. 1.2

b. 0.35 ✓

c. 0.3

d. 0.4

$P(A \text{ and } B) = 0.7 \times 0.5 = 0.35$.

40 [Challenging] • Target: MCQ — Conditional probability • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

If $P(A) = 0.2$, find $P(\text{not } A)$.

a. 0.7

b. 0.9

c. 0.8 ✓

d. 0.2

$P(\text{not } A) = 1 - 0.2 = 0.8$.

41 [Challenging] • Target: MCQ — Counting: permutations and combinations • Cross-Lesson: 6.01 Limits & Derivatives

A data set has variance 100. State its standard deviation.

a. 9

b. 11

c. 100

d. 10 ✓

$\sigma = \sqrt{100} = 10$.

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42 [Challenging] • Target: MCQ — The five-number summary • Cross-Lesson: 7.01 Integration & Applications

A box has 10 balls: 4 red and 6 blue. Two balls are drawn without replacement. Find $P(\text{2nd is red} \mid \text{1st is red})$.

a. $\frac{4}{9}$

b. $\frac{2}{3}$

c. $\frac{1}{3}$ ✓

d. $\frac{2}{5}$

After removing one red ball, 3 of the remaining 9 balls are red: $P = \frac{3}{9} = \frac{1}{3}$.

43 [Challenging] • Target: MCQ — Reading data from a bar chart • Cross-Lesson: 8.01 Exponentials & Logarithms

How many ways can a committee of 2 be chosen from 6 people (order doesn't matter)?

a. 17

b. 30

c. 13

d. 15 ✓

$\binom{6}{2} = \frac{6!}{2!\,(4)!} = 15$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Algebra & Equations

A bag contains 7 red and 4 blue marbles. Find $P(\text{red})$.

a. $\frac{7}{11}$ ✓

b. $\frac{1}{11}$

c. $\frac{4}{11}$

d. $\frac{7}{4}$

7 out of 11 total.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Functions & Transformations

Two independent events have $P(A) = 0.8$ and $P(B) = 0.4$. Find $P(A \text{ and } B)$.

a. 0.37

b. 0.32 ✓

c. 1.2

d. 0.27

$P(A \text{ and } B) = 0.8 \times 0.4 = 0.32$.

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